Pen portraits

A1

This cluster is predominantly composed of UK-born residents and White ethnic groups, with very few Indian residents and no representation of other ethnic groups, including Pakistanis, Black Africans, or other Europeans. The cluster does not exhibit notable concentrations of skilled occupations or high educational qualifications. Housing tenure is mainly social rented, with no flats or detached houses, and some terraces present. This cluster corresponds to the existing A1 cluster in 2001 and 2011, which later evolved into the more ethnically diverse A2 cluster.

A2

Cluster A2 is predominantly composed of UK-born residents, although it has experienced a modest influx of migrants. While heat maps and radar plots do not show any particularly strong distinguishing characteristics, the choropleth maps reveal that A2 has developed over time as a significant cluster, effectively replacing the former A1 cluster. Compared to A1, A2 has seen growth in populations from minority ethnic groups, mainly Black Africans, other Europeans with slightly grown of Indian and Pakistani population. while the proportion of UK-born residents has slightly declined. Housing in A2 shows little change from A1, with few detached houses or flats and some terraces present. Socio-economic characteristics have shifted over time: although values for one-person households and residents aged 0–19 and 20–29 are not prominent in heat maps or radar plots, there has been growth in these age groups and a slight decrease in one-person households compared to A1. Employment in health and education sectors has also increased, reflecting the presence of both UK-born residents and migrant workers, some of whom are students. Educational attainment in the cluster is not strongly linked to Level 3 or Level 4 qualifications. Overall, A2 represents a predominantly UK-born cluster with an increasingly multicultural population, where minority ethnic groups and migrant residents contribute to evolving socio-economic characteristics.

B1

Cluster B1 is a transitional cluster that has largely developed by replacing mainly C1 cluster, reflecting changes in population composition over time. B1 is characterised by a high proportion of migrants born outside the UK, with a predominance of Indian residents and significant growth in Pakistani residents, while the number of UK-born and White ethnic residents has declined. Housing patterns show a reduction in owner occupied tenure and an increase in private rental, indicating both existing population movement and new population influx. Socio-economic and occupational

characteristics, including Level 3 and Level 4 education, business and finance employment, skilled non-manual work, and managerial roles, have remained stable despite the demographic shifts. This suggests that the incoming population is likely related to skilled migration, maintaining the cluster's profile of a highly educated and skilled workforce. Overall, B1 represents a skilled, educated population with increasing ethnic diversity and dynamic population turnover.

C1

This cluster is characterised by a high proportion of UK-born residents and White ethnic groups, with a small presence of Indian residents, making it relatively ethnically homogenous. There are few children aged 0–19, and the 20–29 age group and one person households are also underrepresented. The population has relatively high levels of education (Level 4) and is employed in skilled non-manual occupations, including finance, business, and administration, while student numbers are low. Housing tenure is predominantly owner-occupied, with detached houses being the most common type and very few terraces or flats, and no social rented housing. Overall, this cluster represents a low deprivation, less ethnically diverse population with stable socio-economic characteristics.

D1

Cluster D1 has remained relatively stable over time, with a scattered distribution extending to some outer areas of the city. It is a highly ethnic diverse cluster, with a comparatively large population of Black African residents, as well as migrants born outside the UK, Indian residents, and other European-born citizens. The cluster has the highest concentration of flats and no terraces or detached houses, reflecting its central urban location near the city centre. UK-born residents are comparatively low, and there are no notable concentrations of high-level education, managerial, or professional occupations. D1 is also characterised by a high number of one-person households and 36 relatively few children aged 0–19. Overall, the cluster represents a diverse, urban population dominated by flats and single-person households.

E1

Cluster E1 is characterised by a predominantly Indian population and some Pakistani population , with very low representation of other ethnic groups including white ethic group; also, minimal UK-born residents- making it relatively ethnically homogenous. The cluster has remained temporally stable, showing no significant changes over time. There are no notable concentrations of high educational attainment or skilled occupations. Housing in E2 is dominated by terraced houses, with no flats or detached houses, and tenure is primarily owner-occupied, with very little social rented housing.

One-person households are rare, while residents aged 0–19 low and 20–29 are present in moderate numbers links the characteristics to family households with majority of Indian ethic population.

F1

Cluster F1 is predominantly composed of UK-born residents and White ethnic groups, with a smaller presence of Indian and other European populations. Unlike A1, which is strongly associated with migrant students, F1 includes a higher proportion of residents aged 20–29 and residents are employed in finance, business, and administration, as well as skilled non-manual occupations with some student population , reflecting a cluster of educated and skilled individuals. The cluster is relatively low in ethnic diversity, with UK-born residents remaining dominant. Over time, F1 has not experienced significant growth but has gradually shifted toward the southeast outskirts of the city, with parts of its original area being overtaken by the expanding G1 cluster.

G1

This cluster is characterised by a high concentration of students and residents aged 20 29, with a notable share of one-person households. In contrast, the number of residents aged 0–19 is very low. The cluster is ethnically diverse, including significant proportions of Indian, Black African, and White residents, alongside a large population of international migrants. In particular, residents born outside the UK and from other European countries are strongly represented, while the proportion of UK-born citizens is comparatively low. Geographically, this cluster is located in the city centre, where the built environment is dominated by flats, with no representation of detached houses. 37 Educational Level 1 and Level 3 qualifications population high, reflecting the predominance of students, many of whom are international. Employment-related variables—such as managerial and professional roles, as well as jobs in education, health, finance, and business administration—are comparatively low. Instead, a sizeable share of the population reports working 0–15 hours per week, consistent with student employment patterns. Housing tenure is primarily private rented, with very low levels of owner-occupied and socially rented households, further supporting the interpretation of this cluster as being dominated by migrant student populations pursuing higher education.